

ALL STREAMLIT FUNCTIONS

Pre Requisites

1. OpenAI
 - Environment variable `OPENAI_API_KEY`
2. ElevenLabs
 - Environment variable `ELEVENLABS_API_KEY` with text-to-audio permissions
 - Dubbing voice ids in `.env` file
3. Google
 - Google credentials in repo file `.streamlit/secrets.toml` file
4. `conda`

Installation

To install the required dependencies, run:

```
pip install -r requirements.txt
```

or

```
> conda env create -f conda.yml  
> conda activate sr
```

Running the UI

To start the application, execute:

```
streamlit run ask_ap.py
```

Asking a Question

To just get answer for a question, execute:

```
python -m src.answer "What is suffering?"  
  
### Code  
  
1. Entry point is `ask_ap.py` file
```

Setup TTS Google Creds

◆ Step 1: Create a New Google Cloud Project

1. Go to: <https://console.cloud.google.com/>
 2. In the top navigation bar, click the **project dropdown**.
 3. Click **"New Project"**
 4. Fill in:
 - **Project Name**: e.g., **gtts-assistant** or any name you prefer
 - **Billing Account**: Select an existing billing account, or create one
 - **Organization**: Optional (default is fine)
 5. Click **Create**
 6. Once created, **switch to this new project** using the same dropdown
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◆ Step 2: Enable Required APIs

Enable the **Text-to-Speech API**:

- Navigate to: **APIs & Services** → **Library**
 - Search: **Text-to-Speech API**
 - Click on it and press **Enable**
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◆ Step 3: Create a Service Account

1. Go to: **IAM & Admin** → **Service Accounts**
 2. Click **"Create Service Account"**
 3. Enter the following:
 - **Service account name**: **deployment** (or any name)
 4. Click **Create and Continue**
 5. Assign a role:
 - Recommended: **Project > Editor**
 - Or a more specific role like: **Text-to-Speech Admin**
 6. Click **Done**
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◆ Step 4: Generate and Download Service Account Key

1. In the service account list, click the name of your newly created account
2. Go to the **Keys** tab
3. Click **"Add Key" → "Create new key"**
4. Select **JSON** format
5. Click **Create**

 A **.json** key file will download to your system — keep this file **secure**.

◆ Step 5: Extract Required Fields for Configuration

Open the `.json` key file you downloaded. It will look something like this:

```
{
  "type": "service_account",
  "project_id": "your-project-id",
  "private_key_id": "xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx",
  "private_key": "-----BEGIN PRIVATE KEY-----\nMIIEv...\n-----END\nPRIVATE KEY-----\n",
  "client_email": "deployment@your-project.iam.gserviceaccount.com",
  ...
}
```

◆ Step 6: Format the `[GOOGLE_CREDENTIALS]` Block

Create a block in your configuration file (`.toml`, `.env`, etc.) like this:

```
[GOOGLE_CREDENTIALS]
type = "service_account"
project_id = "your-project-id"
private_key_id = "your-private-key-id"
private_key = """
-----BEGIN PRIVATE KEY-----
...paste the full key here with newlines...
-----END PRIVATE KEY-----
"""

client_email = "deployment@your-project.iam.gserviceaccount.com"
client_id = "your-client-id"
auth_uri = "https://accounts.google.com/o/oauth2/auth"
token_uri = "https://oauth2.googleapis.com/token"
auth_provider_x509_cert_url =
"https://www.googleapis.com/oauth2/v1/certs"
client_x509_cert_url =
"https://www.googleapis.com/robot/v1/metadata/x509/deployment@your-
project.iam.gserviceaccount.com"
universe_domain = "googleapis.com"
```

⚠ **Important:** Use triple quotes (`"""..."""`) for the private key to handle newlines properly.